# Trustworthy ML

## Generalisation Studies



Please scan
before we start
and keep the tab open

**Antonia Marcu**

# Would you trust a machine learning model with your medical diagnosis?



# Some evidence for

Liu, Xiaoxuan, et al. "A comparison of deep learning performance against health-care professionals in detecting diseases from medical imaging: a systematic review and meta-analysis." *The lancet digital health* 1.6 (2019): e271-e297.

What would you ask the ML developers to feel confident in the model they trained?

- What type of model did you use?
- How many training and test points did you have?
- How many samples did you have for each class?
- What was your training setup?
- What is the test accuracy of your model?

# When exactly can we say a model generalises?

# Generalisation

## Quick Recap



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Not so clear

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# When exactly can we say a model generalises?

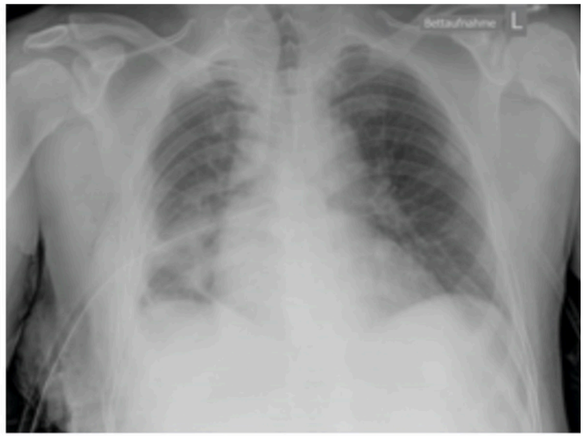
Not so clear

- Comparable to human accuracy? Sometimes machines do better. How much better?
- Why is this problematic? (Think e.g. regulations)

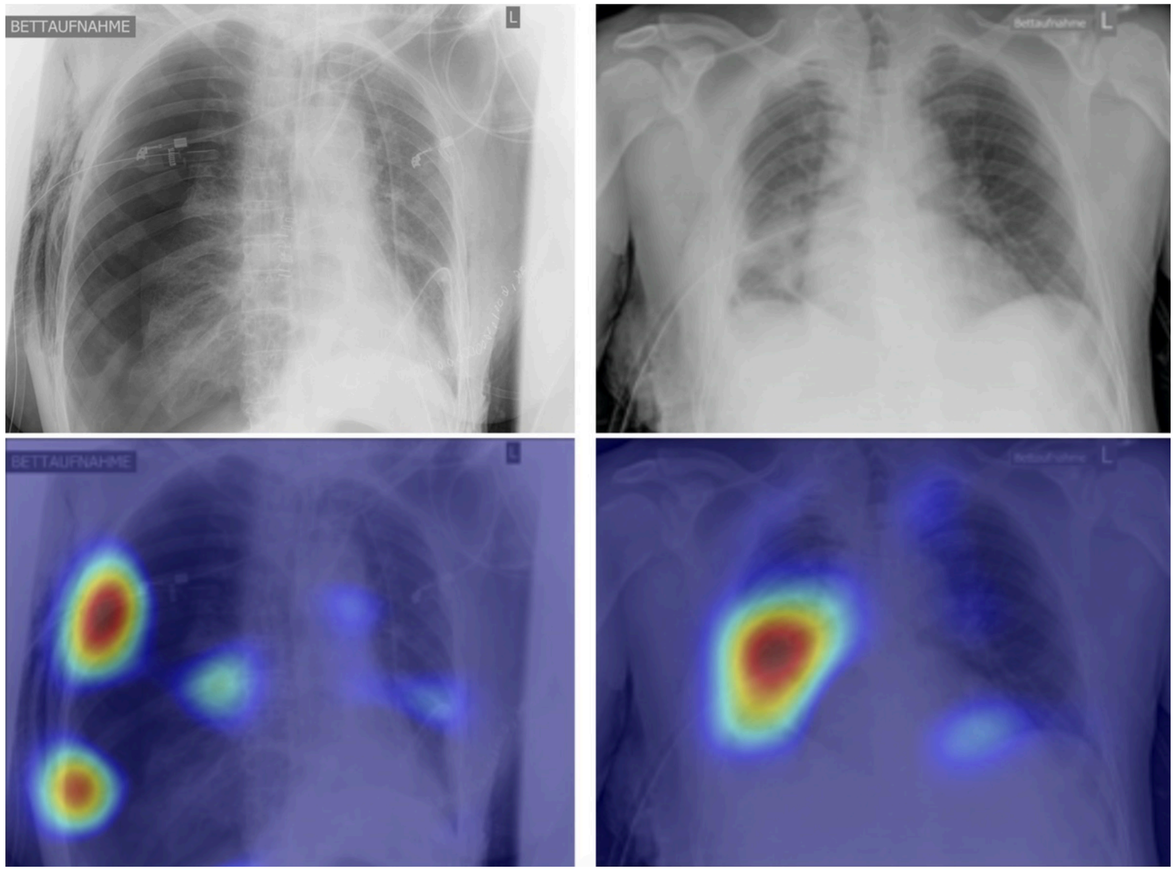
Let’s assume we have a clear notion of “good enough” accuracy

# Why Accuracy is Not All We Should Care About

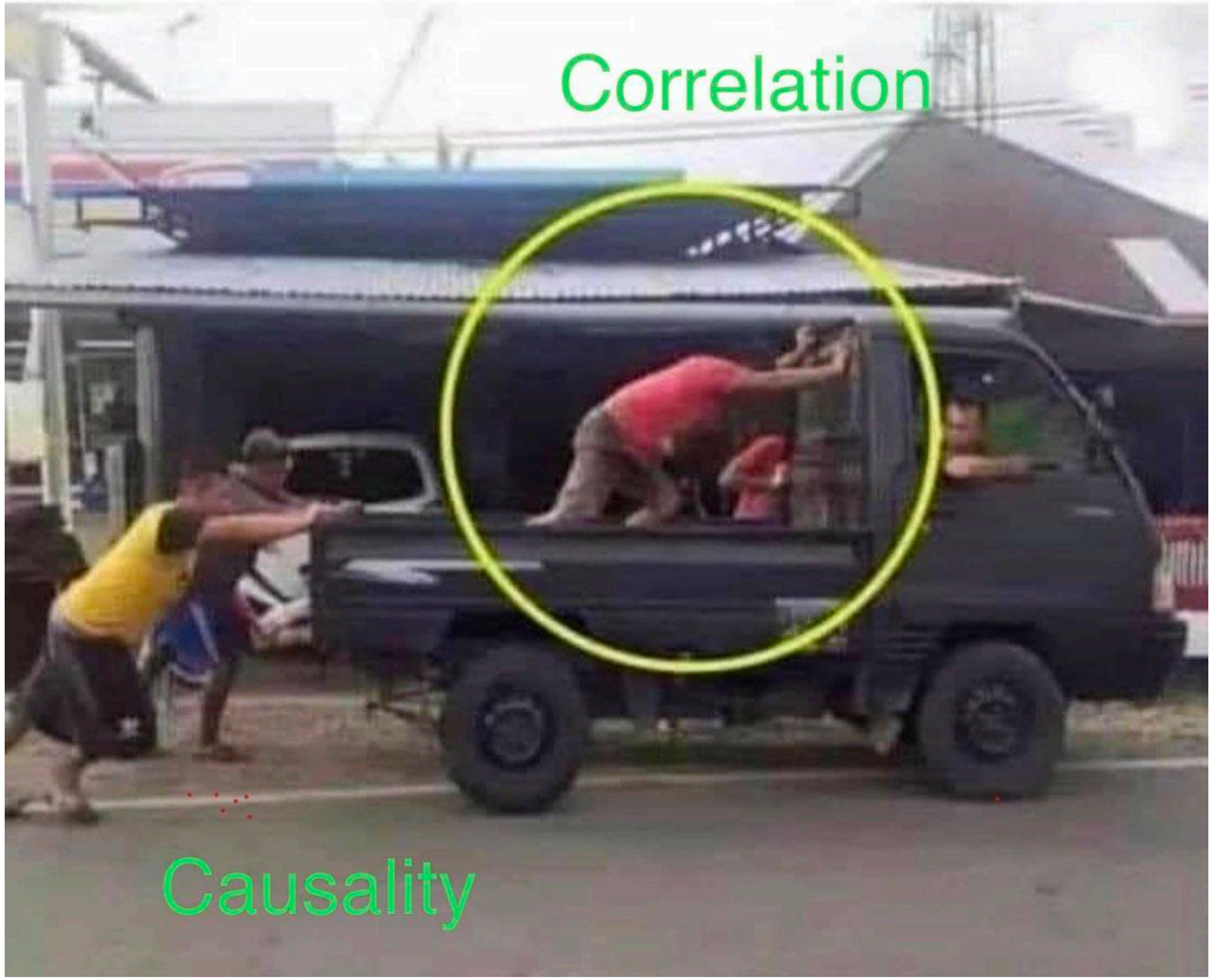
- What factors would you consider when collecting data for an automated chest scan diagnosis?



# Why Accuracy is Not All We Should Care About



# Spurious Correlations



What can you do to verify that you are not accidentally doing the same thing?

# Practical Stuff You Could Consider

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    2. Confounding factors

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    2. Confounding factors

Where might you hope to see that the pattern used by the model is missing in a new set of data?

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* let’s keep this in mind for a second

(Reminder: Let’s assume we have a clear notion of “good enough” accuracy)
What else should we practically evaluate in a model?

# Which one is the most dangerous to predict?

- True positive, True negative, False positive, False negative

# How can we make sure we don’t have false negatives?

# Outputs as Probabilities

## Quick Recap

Input sample



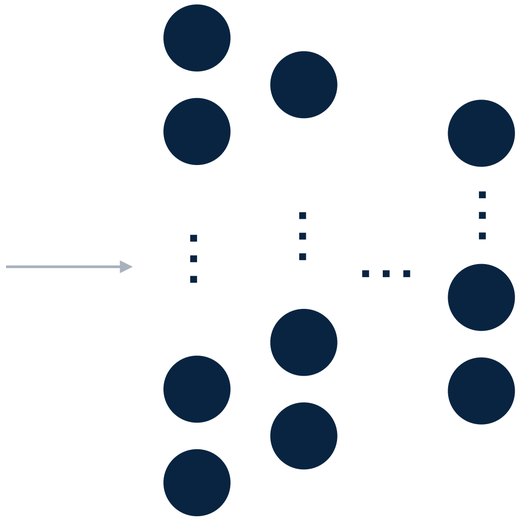
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## Quick Recap

Input sample

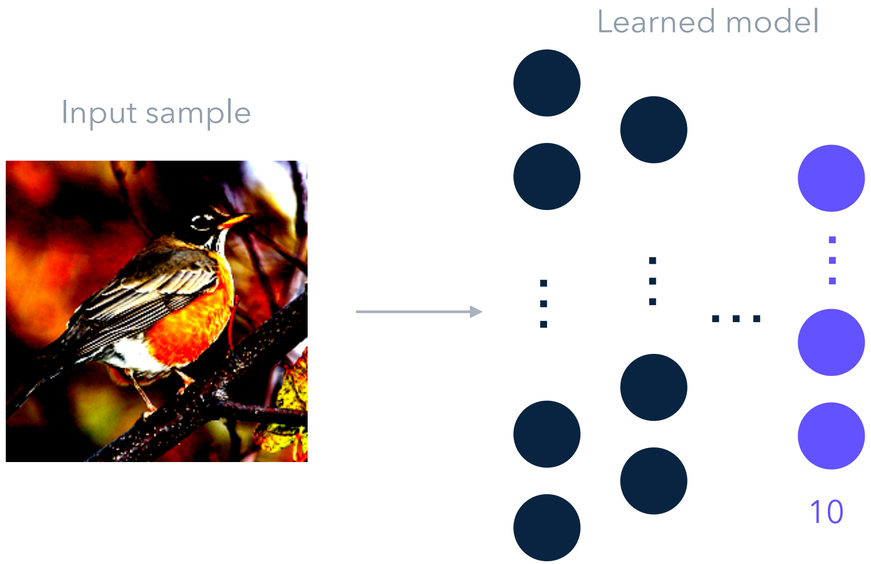


Learned model



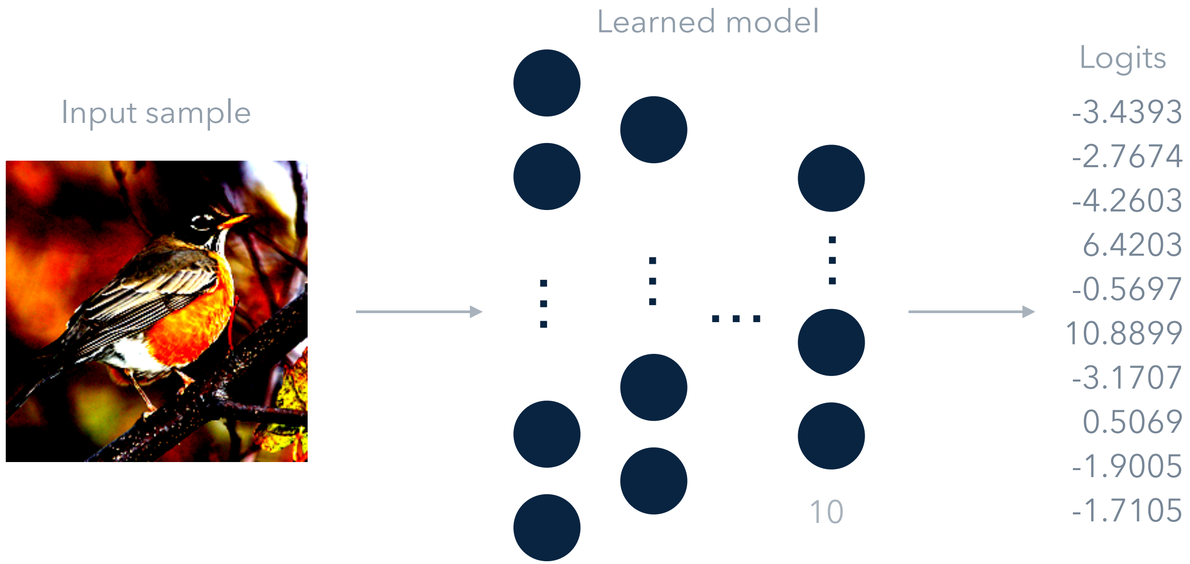
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## Quick Recap



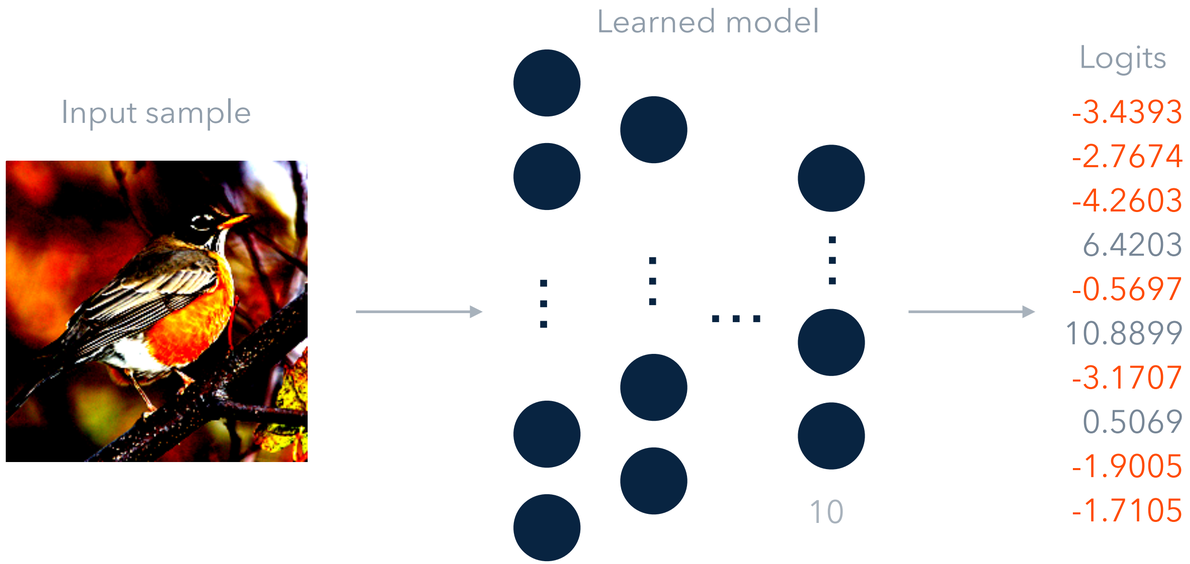
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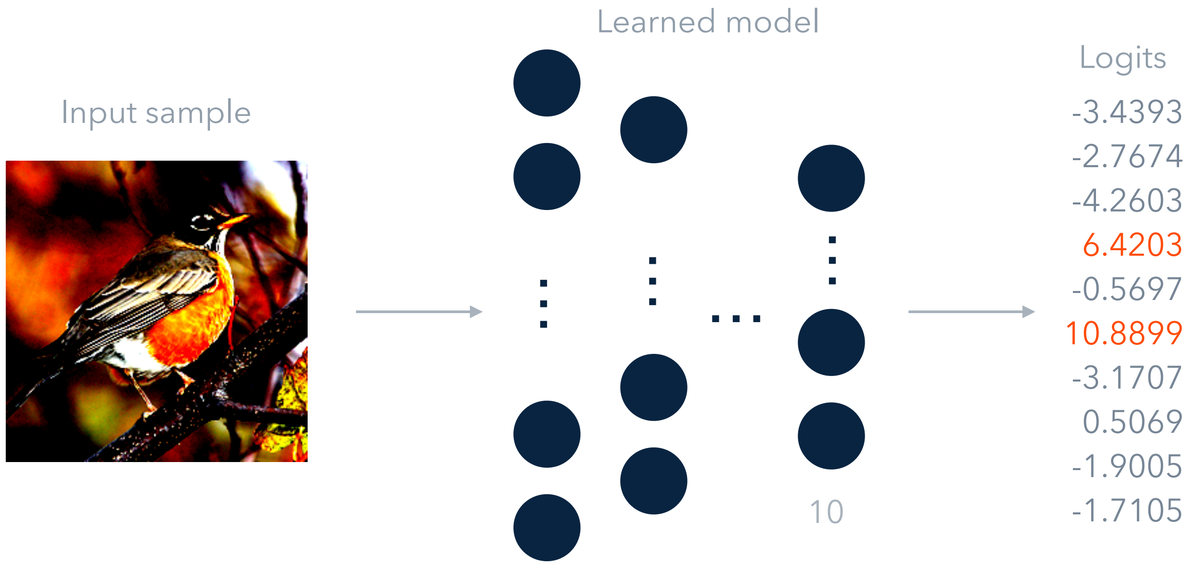
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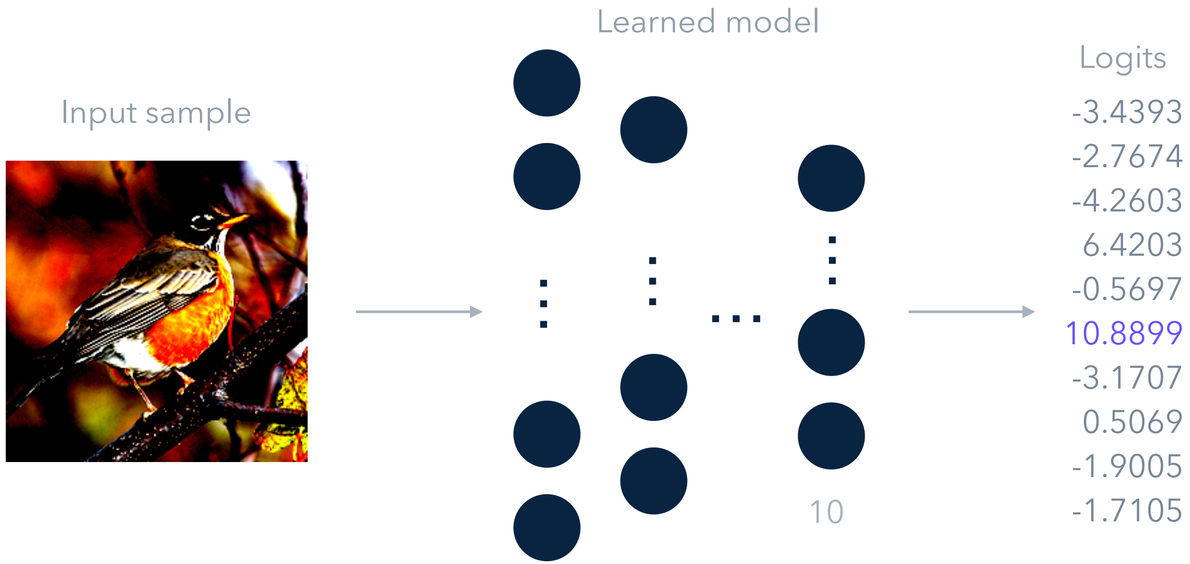
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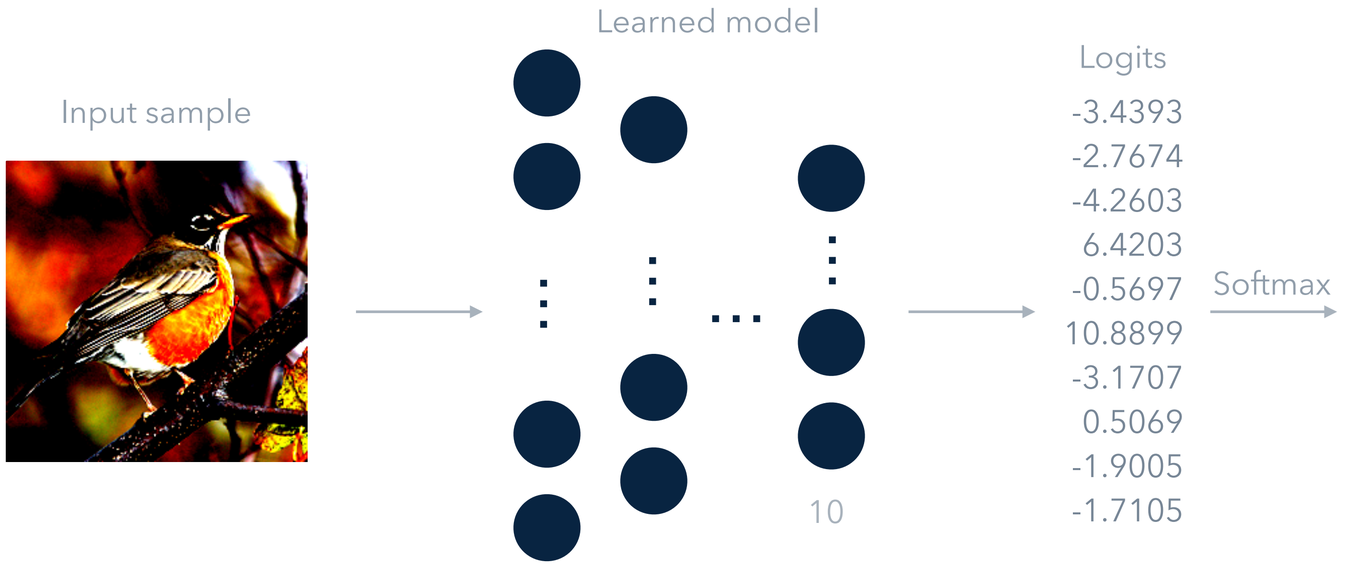
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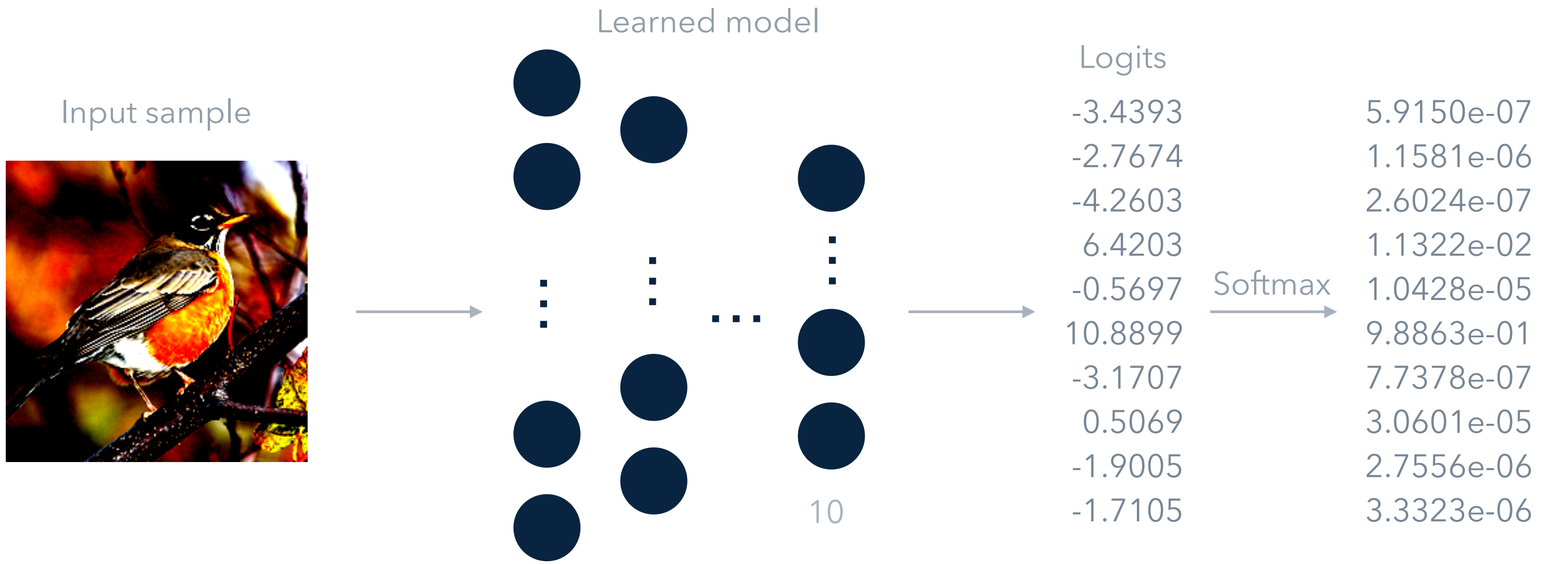
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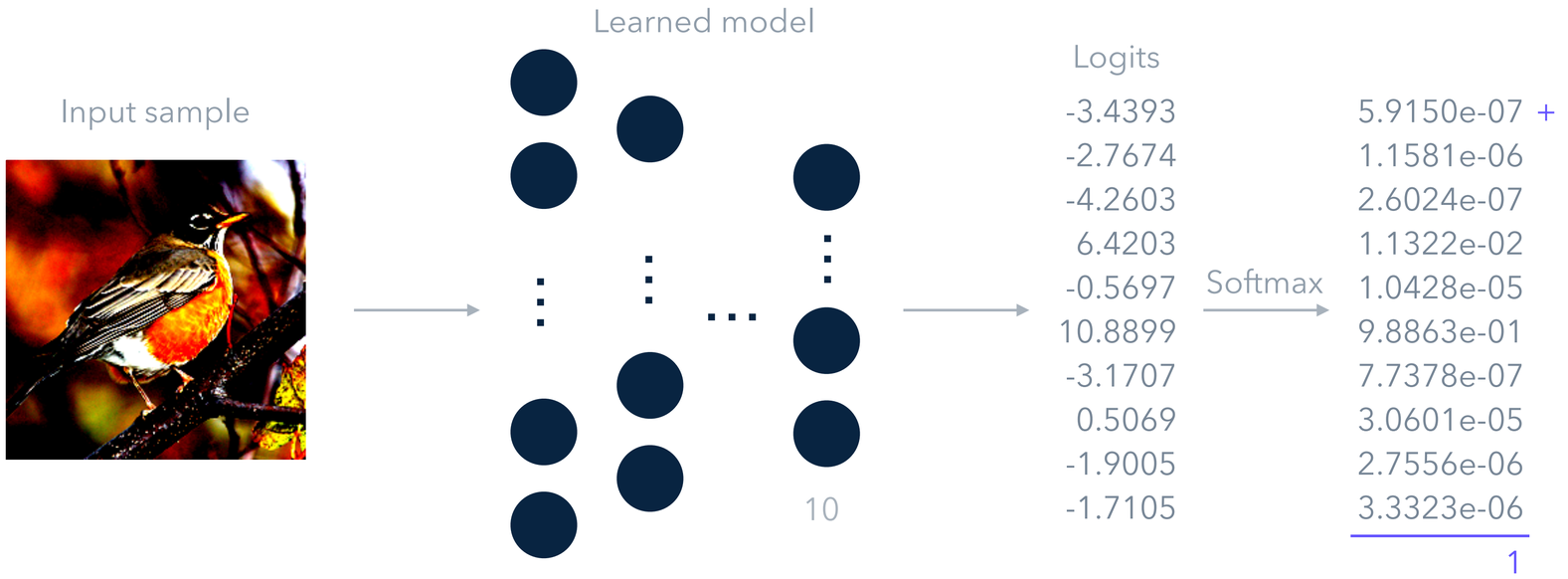
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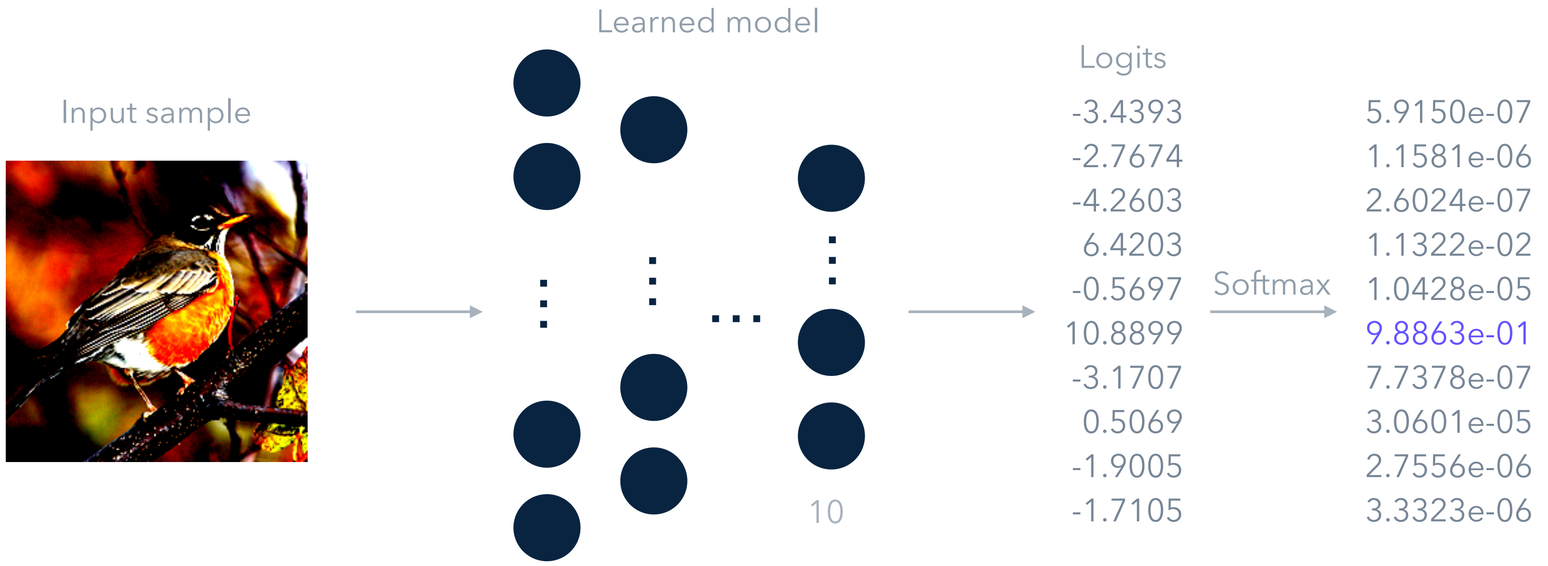
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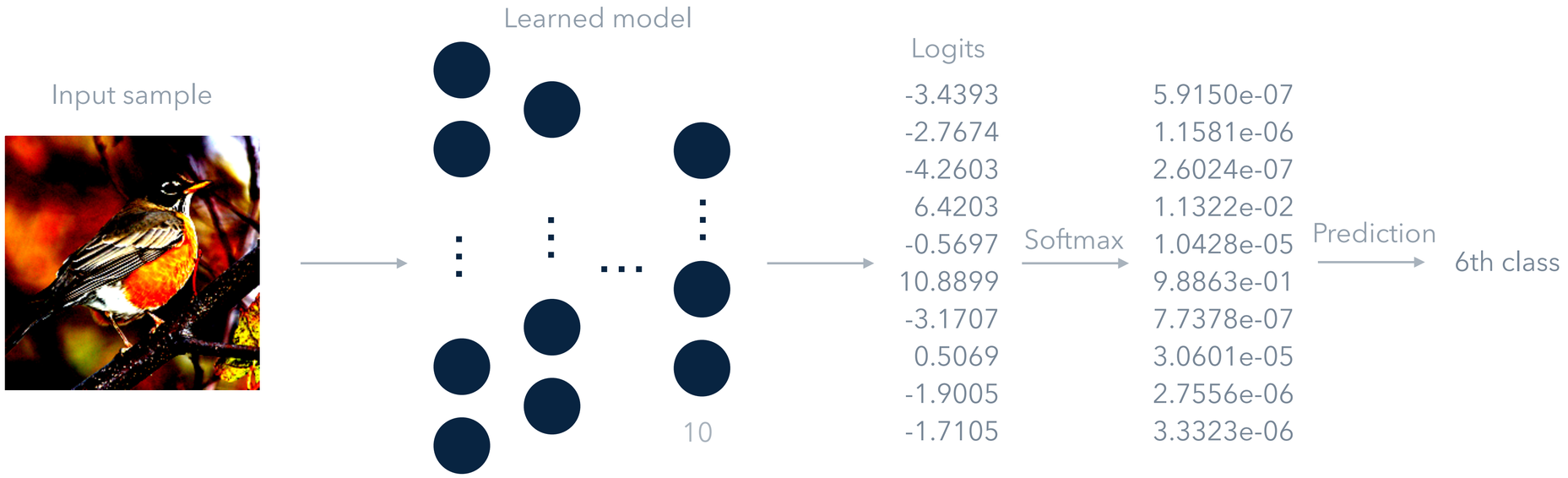
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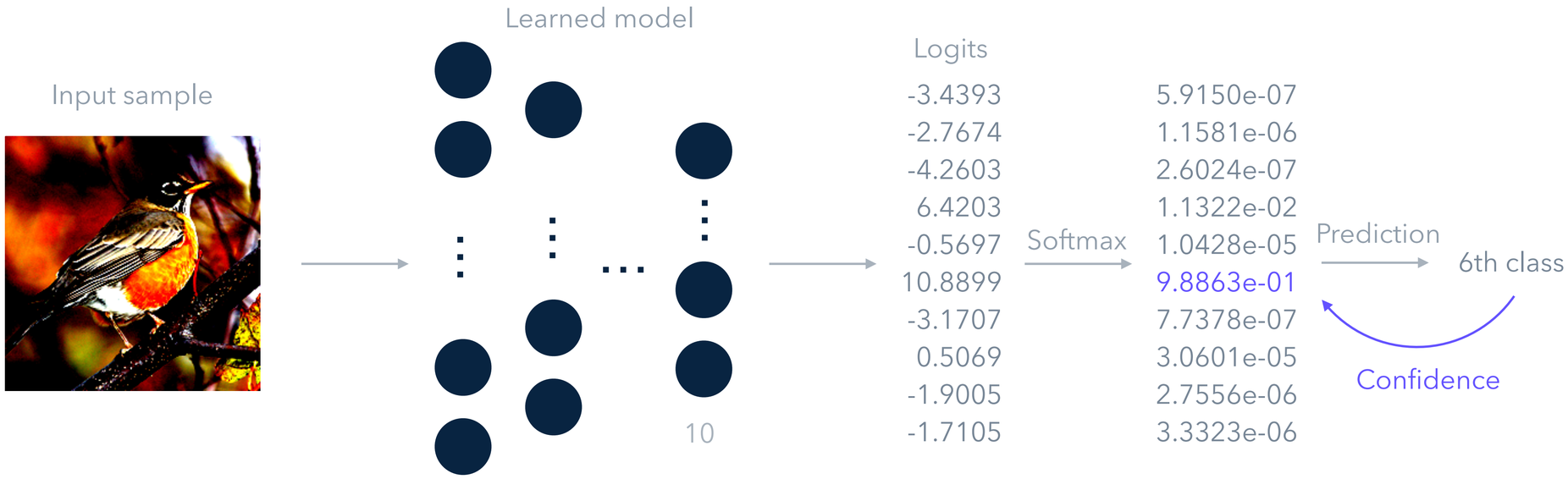
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## Quick Recap



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## Quick Recap



# Reminder: How can we make sure we don’t have false negatives?

# Threshold on confidence?

# Calibration

- Deep learning models are very confident in their predictions

# Calibration

- Deep learning models are very confident in their predictions even when the predictions are incorrect

# Calibration

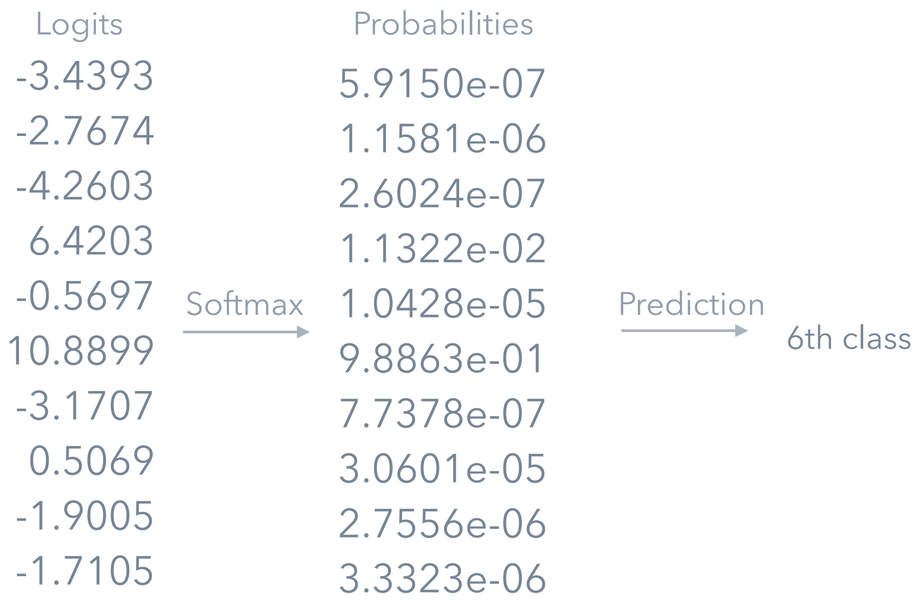
- Deep learning models are very confident in their predictions even when the predictions are incorrect
- This is really problematic

# Calibration

- Deep learning models are very confident in their predictions even when the predictions are incorrect
- This is really problematic
  - Practically:
    - We can’t use it as a filtering step
    - We can’t use it to signal reliance on an unaligned pattern
  - Theoretically, they can both boil down to distributions shifts

# But why are models overconfident?

# Going Back


Logits
-3.4393
-2.7674
-4.2603
6.4203
-0.5697
10.8899
-3.1707
0.5069
-1.9005
-1.7105
Softmax
Probabilities
5.9150e-07
1.1581e-06
2.6024e-07
1.1322e-02
1.0428e-05
9.8863e-01
7.7378e-07
3.0601e-05
2.7556e-06
3.3323e-06
Prediction
6th class

# Going Back

Let’s assume

| Target | Logits | | Probabilities | | |
|---|---|---|---|---|---|
| 0 | -3.4393 | | 5.9150e-07 | | |
| 0 | -2.7674 | | 1.1581e-06 | | |
| 0 | -4.2603 | | 2.6024e-07 | | |
| 0 | 6.4203 | | 1.1322e-02 | | |
| 0 | -0.5697 | Softmax → | 1.0428e-05 | Prediction → | 6th class |
| 1 | 10.8899 | | 9.8863e-01 | | |
| 0 | -3.1707 | | 7.7378e-07 | | |
| 0 | 0.5069 | | 3.0601e-05 | | |
| 0 | -1.9005 | | 2.7556e-06 | | |
| 0 | -1.7105 | | 3.3323e-06 | | |

# Cross Entropy Loss

| Target | Logits |
|---|---|
| 0 | -3.4393 |
| 0 | -2.7674 |
| 0 | -4.2603 |
| 0 | 6.4203 |
| 0 | -0.5697 |
| 1 | 10.8899 |
| 0 | -3.1707 |
| 0 | 0.5069 |
| 0 | -1.9005 |
| 0 | -1.7105 |

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| Target | Logits |
|---|---|
| 0 | -3.4393 |
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| 0 | -4.2603 |
| 0 | 6.4203 |
| 0 | -0.5697 |
| 1 | 10.8899 |
| 0 | -3.1707 |
| 0 | 0.5069 |
| 0 | -1.9005 |
| 0 | -1.7105 |

Cross Entropy Loss → 0.0114

# Cross Entropy Loss

| Target | Logits |
|---|---|
| 0 | -3.4393 |
| 0 | -2.7674 |
| 0 | -4.2603 |
| 0 | 6.4203 |
| 0 | -0.5697 |
| 1 | 10.8899 |
| 0 | -3.1707 |
| 0 | 0.5069 |
| 0 | -1.9005 |
| 0 | -1.7105 |

Cross Entropy → Loss

0.0114 ↓

# Cross Entropy Loss

| Target | Logits |
|---|---|
| 0 | -3.4393 |
| 0 | -2.7674 |
| 0 | -4.2603 |
| 0 | 6.4203 |
| 0 | -0.5697 |
| 1 | 10.8899 |
| 0 | -3.1707 |
| 0 | 0.5069 |
| 0 | -1.9005 |
| 0 | -1.7105 |

Cross Entropy Loss → 0.0114

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| Target | Logits |
|---|---|
| 0 | -3.4393 |
| 0 | -2.7674 |
| 0 | -4.2603 |
| 0 | 6.4203 |
| 0 | -0.5697 |
| 1 | 11.8899 |
| 0 | -3.1707 |
| 0 | 0.5069 |
| 0 | -1.9005 |
| 0 | -1.7105 |

Cross Entropy Loss → 0.0042

# Cross Entropy Loss

| Target | Logits |
|---|---|
| 0 | -3.4393 |
| 0 | -2.7674 |
| 0 | -4.2603 |
| 0 | 6.4203 |
| 0 | -0.5697 |
| 1 | 12.8899 |
| 0 | -3.1707 |
| 0 | 0.5069 |
| 0 | -1.9005 |
| 0 | -1.7105 |

Cross Entropy Loss → 0.0016

# Cross Entropy Loss

| Target | Logits |
|---|---|
| 0 | -3.4393 |
| 0 | -2.7674 |
| 0 | -4.2603 |
| 0 | 6.4203 |
| 0 | -0.5697 |
| 1 | 13.8899 |
| 0 | -3.1707 |
| 0 | 0.5069 |
| 0 | -1.9005 |
| 0 | -1.7105 |

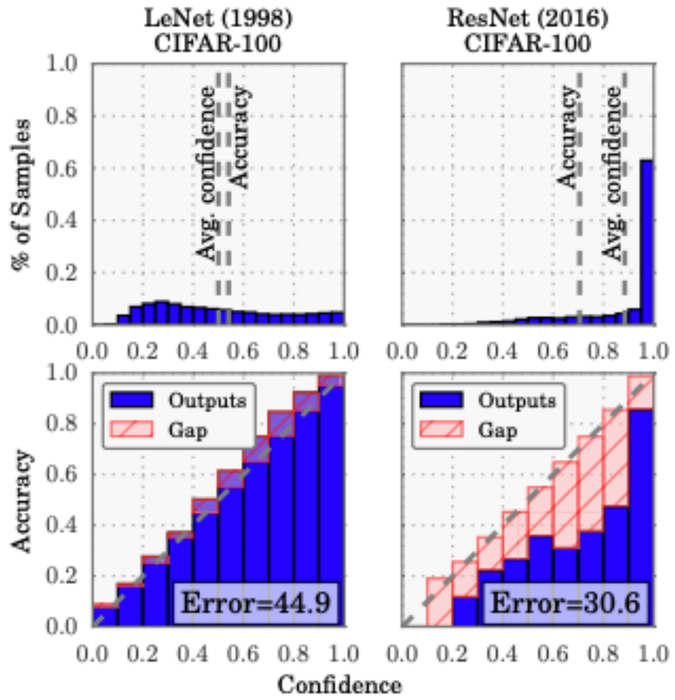
Cross Entropy Loss → 0.0006

# Back to Calibration

- Deep learning models are very confident in their predictions even when the predictions are incorrect
- This is really problematic
- We are currently rewarding models for being overconfident

# Measures of Calibration

$$\text{ECE} = \sum_b^B \frac{n_b}{N} |\text{acc}(b) - \text{conf}(b)|$$




On Calibration of Modern Neural Networks
Calibration in Deep Learning: A Survey of the State-of-the-Art
Measuring Calibration in Deep Learning

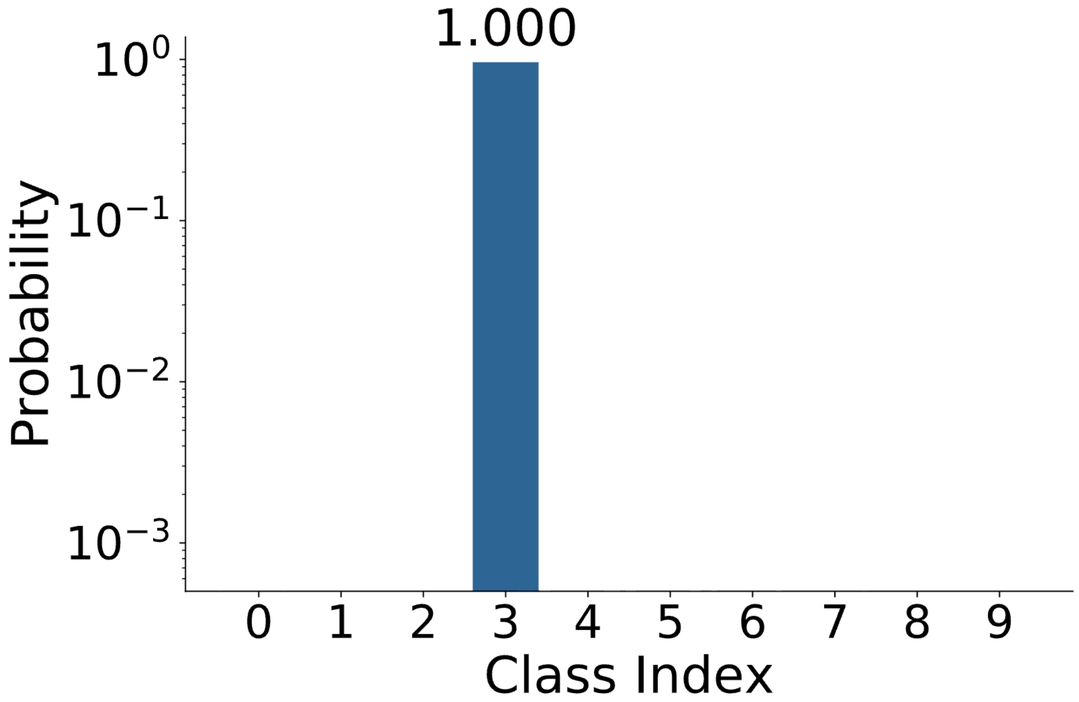
# Attempts to Address Calibration

# Attempts* to Address Calibration

* 1. This is still an open research question
* 2. The observed calibration improvements were usually coincidental

# Attempts to Address Calibration

- Temperature scaling



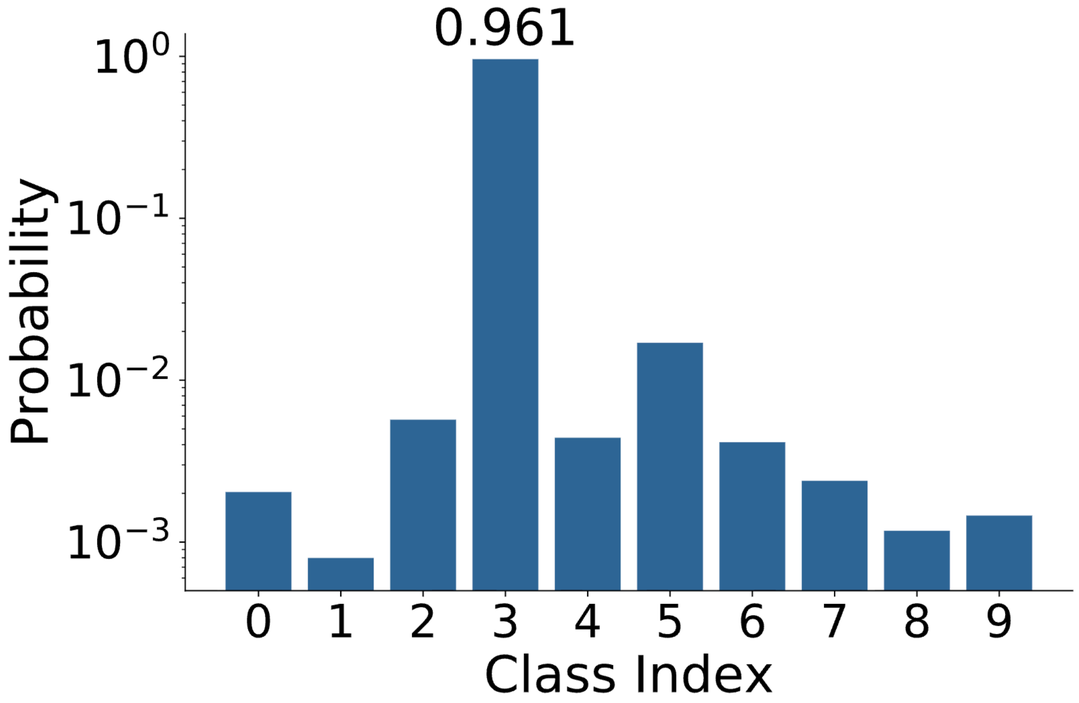
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# What does this not account for?

# Attempts to Address Calibration

- Temperature scaling but it does not account for input-related relationships between classes

# Attempts to Address Calibration

- Temperature scaling
- Mixup

# Mixup





# Mixup

30% *



+ 70% *



# Mixup

30 % dog + 70% * cat



# Mixup

What about the labels?



# What does this not account for?

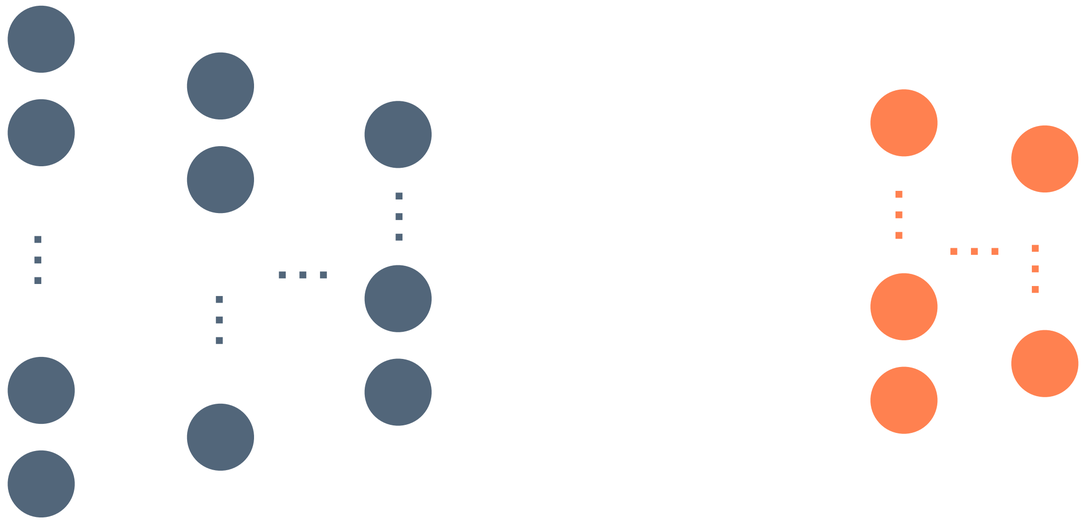
# Attempts to Address Calibration

- Temperature scaling
- Mixup but it does not account for data-driven relationships between classes (i.e. class “similarity”)

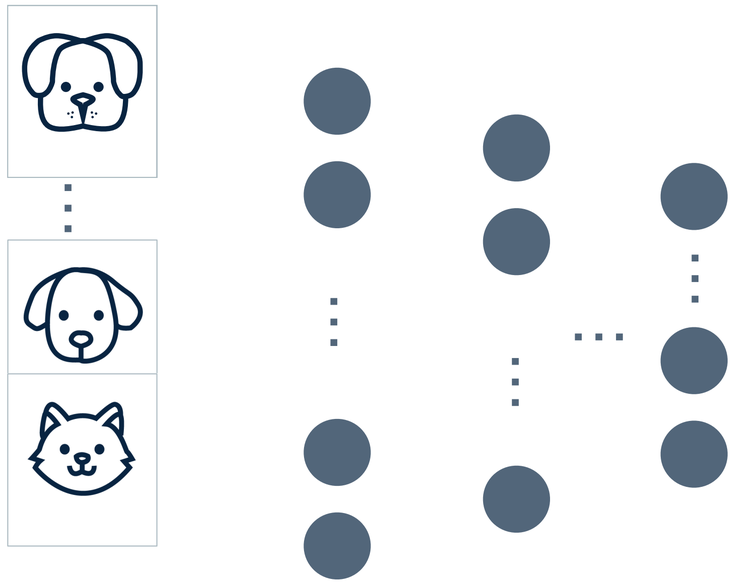
# Attempts to Address Calibration

- Temperature scaling
- Mixup
- Knowledge distillation

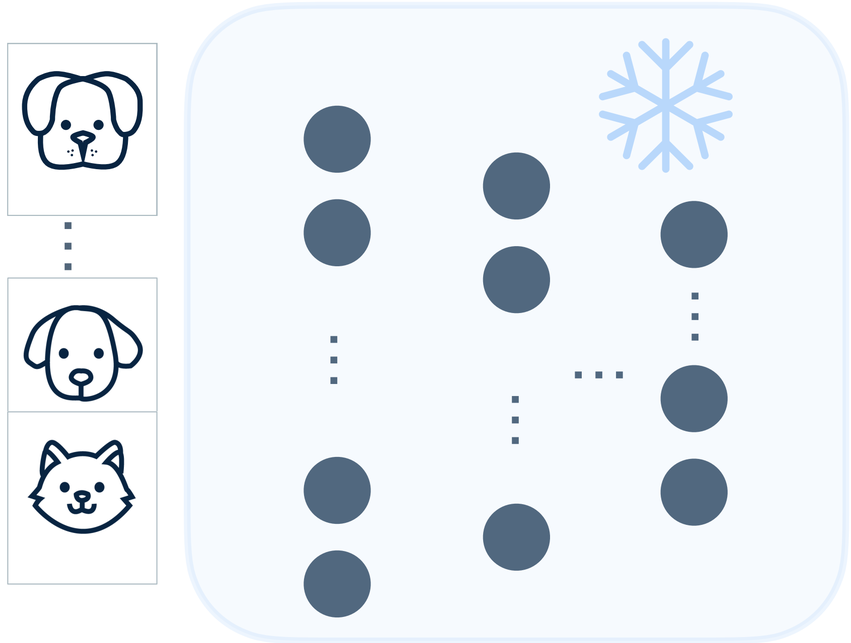
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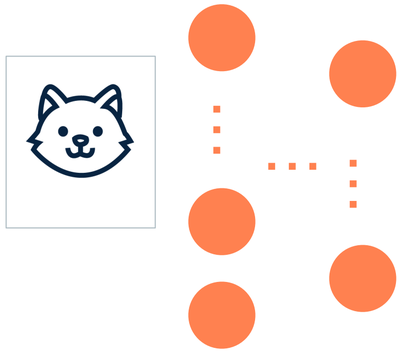
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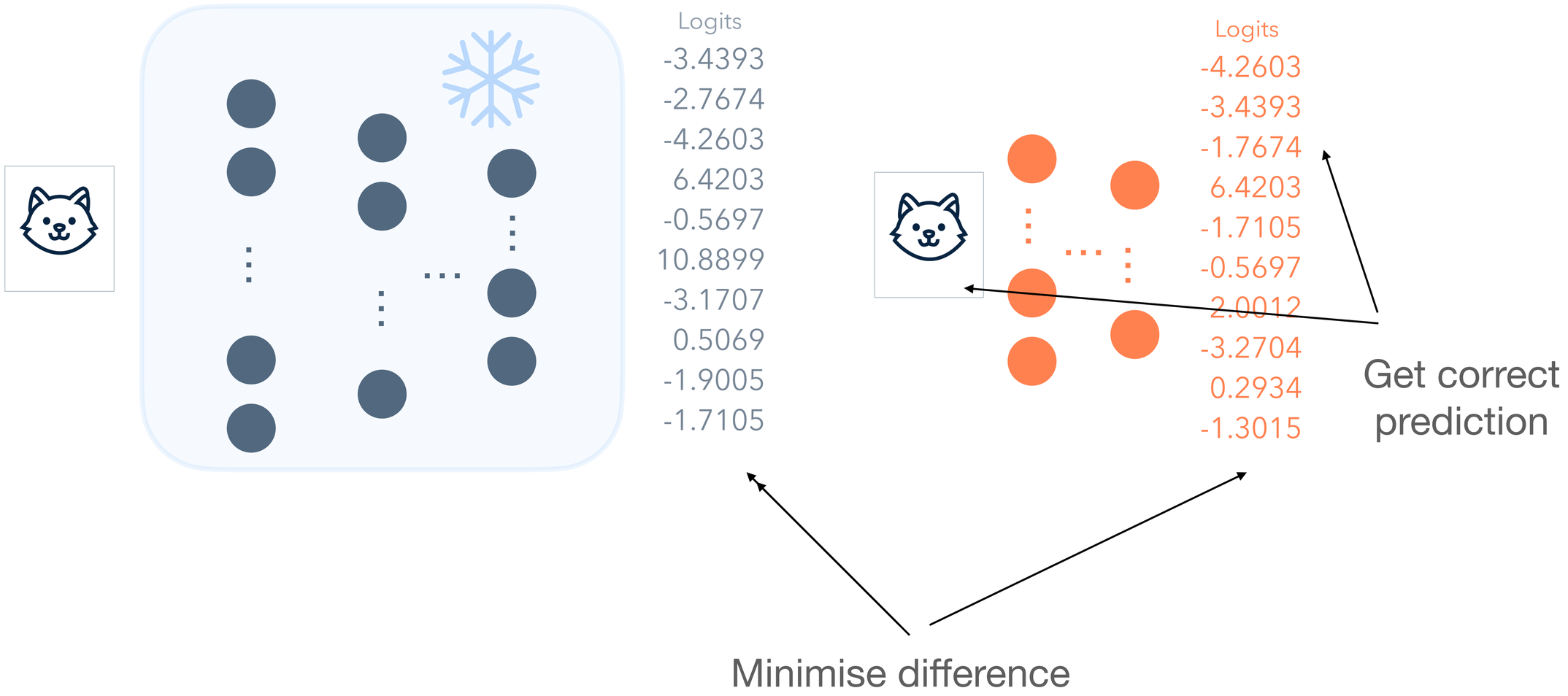


Logits
-3.4393
-2.7674
-4.2603
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-3.1707
0.5069
-1.9005
-1.7105



Logits
-4.2603
-3.4393
-1.7674
6.4203
-1.7105
-0.5697
2.0012
-3.2704
0.2934
-1.3015

# Knowledge Distillation (KD)



# Knowledge Distillation (KD)

- Many variants have been proposed since, but this is the principle
- Typically, the loss becomes a weighted sum of the two objectives

e.g. alpha * KL + (1-alpha) * CE

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- This is yet another open research question